

IT1223 (P) - Database Management Systems

(P) Department of Physical Science

University of Vavuniya

Revision Work Sheet – 03

1. Create the appropriate tables with the suitable structure and insert tuples in it

```
CREATE DATABASE Sales_db;
```

```
USE Sales_db;
```

```
CREATE TABLE salesman (  
SALESMAN_ID NUMERIC(5)  
PRIMARY KEY,  
NAME VARCHAR(30),  
CITY VARCHAR(15),  
COMMISSION DECIMAL(5, 2) );
```

```
CREATE TABLE customer (  
CUSTOMER_ID NUMERIC(5)  
PRIMARY KEY, CUST_NAME VARCHAR(30),  
CITY VARCHAR(15), GRADE NUMERIC(3),  
SALESMAN_ID NUMERIC(5),  
FOREIGN KEY (SALESMAN_ID) REFERENCES salesman(SALESMAN_ID) );
```

```
CREATE TABLE orders (  
ORD_NO NUMERIC(5) PRIMARY KEY,  
PURCH_AMT DECIMAL(8, 2),  
ORD_DATE DATE, CUSTOMER_ID NUMERIC(5),  
SALESMAN_ID NUMERIC(5), FOREIGN KEY (CUSTOMER_ID) REFERENCES customer(CUSTOMER_ID),  
FOREIGN KEY (SALESMAN_ID) REFERENCES salesman(SALESMAN_ID) );
```

```
INSERT INTO salesman VALUES  
(10001, 'John Smith', 'New York', 13.50),  
(10002, 'Jane Doe', 'Los Angeles', 15.00),  
(10003, 'George Brown', 'Chicago', 12.75),  
(10004, 'Nancy Davis', 'Houston', 11.00),  
(10005, 'Peter Wilson', 'Phoenix', 14.25),  
(10006, 'Linda Johnson', 'Philadelphia', 10.50);
```

```
INSERT INTO customer VALUES  
(20001, 'Alpha Corp', 'New York', 1, 10001),
```

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(20002, 'Beta Inc', 'Los Angeles', 2, 10002),
(20003, 'Gamma LLC', 'Chicago', 3, 10003),
(20004, 'Delta Ltd', 'Houston', 1, 10004),
(20005, 'Epsilon SA', 'Phoenix', 2, 10005),
(20006, 'Zeta Pty', 'Philadelphia', 3, 10006);

INSERT INTO orders VALUES

(30001, 500.50, '2024-01-15', 20001, 10001),
(30002, 750.00, '2024-02-20', 20002, 10002),
(30003, 200.75, '2024-03-10', 20003, 10003),
(30004, 950.25, '2024-04-05', 20004, 10004),
(30005, 1200.00, '2024-05-15', 20005, 10005),
(30006, 650.50, '2024-06-20', 20006, 10006);

```
MariaDB [(none)]> CREATE DATABASE Sales_db;  
Query OK, 1 row affected (0.002 sec)
```

```
MariaDB [(none)]> USE Sales_db;  
Database changed  
MariaDB [Sales_db]> USE Sales_db;  
Database changed
```

```
MariaDB [Sales_db]>  
MariaDB [Sales_db]> CREATE TABLE salesman (  
->     SALESMAN_ID NUMERIC(5) PRIMARY KEY,  
->     NAME VARCHAR(30),  
->     CITY VARCHAR(15),  
->     COMMISSION DECIMAL(5, 2)  
-> );  
Query OK, 0 rows affected (0.010 sec)
```

```
MariaDB [Sales_db]> CREATE TABLE customer (  
->     CUSTOMER_ID NUMERIC(5) PRIMARY KEY,  
->     CUST_NAME VARCHAR(30),  
->     CITY VARCHAR(15),  
->     GRADE NUMERIC(3),  
->     SALESMAN_ID NUMERIC(5),  
->     FOREIGN KEY (SALESMAN_ID) REFERENCES salesman(SALESMAN_I  
-> );  
Query OK, 0 rows affected (0.015 sec)
```

```
MariaDB [Sales_db]>  
MariaDB [Sales_db]> CREATE TABLE orders (  
->     ORD_NO NUMERIC(5) PRIMARY KEY,  
->     PURCH_AMT DECIMAL(8, 2),  
->     ORD_DATE DATE,  
->     CUSTOMER_ID NUMERIC(5),  
->     SALESMAN_ID NUMERIC(5),  
->     FOREIGN KEY (CUSTOMER_ID) REFERENCES customer(CUSTOMER_I  
->     FOREIGN KEY (SALESMAN_ID) REFERENCES salesman(SALESMAN_I  
-> );  
Query OK, 0 rows affected (0.018 sec)
```

```
MariaDB [Sales_db]> INSERT INTO salesman VALUES  
-> (10001, 'John Smith', 'New York', 13.50),  
-> (10002, 'Jane Doe', 'Los Angeles', 15.00),  
-> (10003, 'George Brown', 'Chicago', 12.75),  
-> (10004, 'Nancy Davis', 'Houston', 11.00),  
-> (10005, 'Peter Wilson', 'Phoenix', 14.25),  
-> (10006, 'Linda Johnson', 'Philadelphia', 10.50);  
Query OK, 6 rows affected (0.015 sec)
```

```
MariaDB [Sales_db]> INSERT INTO customer VALUES
-> (20001, 'Alpha Corp', 'New York', 1, 10001),
-> (20002, 'Beta Inc', 'Los Angeles', 2, 10002),
-> (20003, 'Gamma LLC', 'Chicago', 3, 10003),
-> (20004, 'Delta Ltd', 'Houston', 1, 10004),
-> (20005, 'Epsilon SA', 'Phoenix', 2, 10005),
-> (20006, 'Zeta Pty', 'Philadelphia', 3, 10006);
Query OK, 6 rows affected (0.007 sec)
Records: 6 Duplicates: 0 Warnings: 0

MariaDB [Sales_db]> INSERT INTO orders VALUES
-> (30001, 500.50, '2024-01-15', 20001, 10001),
-> (30002, 750.00, '2024-02-20', 20002, 10002),
-> (30003, 200.75, '2024-03-10', 20003, 10003),
-> (30004, 950.25, '2024-04-05', 20004, 10004),
-> (30005, 1200.00, '2024-05-15', 20005, 10005),
-> (30006, 650.50, '2024-06-20', 20006, 10006);
Query OK, 6 rows affected (0.006 sec)
Records: 6 Duplicates: 0 Warnings: 0
```

2.Show the structure of all the tables:

```
DESCRIBE salesman;
DESCRIBE customer;
DESCRIBE orders;
```

```
MariaDB [Sales_db]> DESCRIBE salesman;
```

Field	Type	Null	Key	Default	Extra
SALESMAN_ID	decimal(5,0)	NO	PRI	NULL	
NAME	varchar(30)	YES		NULL	
CITY	varchar(15)	YES		NULL	
COMMISSION	decimal(5,2)	YES		NULL	

```
4 rows in set (0.008 sec)
```

```
MariaDB [Sales_db]> DESCRIBE customer;
```

Field	Type	Null	Key	Default	Extra
CUSTOMER_ID	decimal(5,0)	NO	PRI	NULL	
CUST_NAME	varchar(30)	YES		NULL	
CITY	varchar(15)	YES		NULL	
GRADE	decimal(3,0)	YES		NULL	
SALESMAN_ID	decimal(5,0)	YES	MUL	NULL	

```
5 rows in set (0.008 sec)
```

```
MariaDB [Sales_db]> DESCRIBE orders;
```

Field	Type	Null	Key	Default	Extra
ORD_NO	decimal(5,0)	NO	PRI	NULL	
PURCH_AMT	decimal(8,2)	YES		NULL	
ORD_DATE	date	YES		NULL	
CUSTOMER_ID	decimal(5,0)	YES	MUL	NULL	
SALESMAN_ID	decimal(5,0)	YES	MUL	NULL	

```
5 rows in set (0.009 sec)
```

3.Display the CUST_NAME and GRADE of all the customers:

```
SELECT CUST_NAME, GRADE FROM customer;
```

```
MariaDB [Sales_db]> SELECT CUST_NAME, GRADE FROM customer;
```

CUST_NAME	GRADE
Alpha Corp	1
Beta Inc	2
Gamma LLC	3
Delta Ltd	1
Epsilon SA	2
Zeta Pty	3

```
6 rows in set (0.001 sec)
```

4. Retrieve the NAME and CITY columns from the salesman table where COMMISSION is greater than 12:

```
SELECT NAME, CITY FROM salesman WHERE COMMISSION > 12;
```

```
MariaDB [Sales_db]> SELECT NAME, CITY FROM salesman WHERE COMMISSION > 12;
```

NAME	CITY
John Smith	New York
Jane Doe	Los Angeles
George Brown	Chicago
Peter Wilson	Phoenix

```
4 rows in set (0.002 sec)
```

5. Select all columns from the orders table, ordered by PURCH_AMT in descending order:

```
SELECT * FROM orders ORDER BY PURCH_AMT DESC;
```

```
MariaDB [Sales_db]> SELECT * FROM orders ORDER BY PURCH_AMT DESC;
+-----+-----+-----+-----+-----+
| ORD_NO | PURCH_AMT | ORD_DATE | CUSTOMER_ID | SALESMAN_ID |
+-----+-----+-----+-----+-----+
| 30005 | 1200.00 | 2024-05-15 | 20005 | 10005 |
| 30004 | 950.25 | 2024-04-05 | 20004 | 10004 |
| 30002 | 750.00 | 2024-02-20 | 20002 | 10002 |
| 30006 | 650.50 | 2024-06-20 | 20006 | 10006 |
| 30001 | 500.50 | 2024-01-15 | 20001 | 10001 |
| 30003 | 200.75 | 2024-03-10 | 20003 | 10003 |
+-----+-----+-----+-----+-----+
6 rows in set (0.001 sec)
```

6. Display the details of all the salesmen in 'New York' or with a COMMISSION greater than 14:

```
SELECT * FROM salesman WHERE CITY = 'New York' OR COMMISSION > 14;
```

```
MariaDB [Sales_db]> SELECT * FROM salesman WHERE CITY = 'New York' OR COMMISSION > 14;
+-----+-----+-----+-----+
| SALESMAN_ID | NAME | CITY | COMMISSION |
+-----+-----+-----+-----+
| 10001 | John Smith | New York | 13.50 |
| 10002 | Jane Doe | Los Angeles | 15.00 |
| 10005 | Peter Wilson | Phoenix | 14.25 |
+-----+-----+-----+-----+
3 rows in set (0.001 sec)
```

7. Select all customers with a GRADE of 1 and not located in 'Chicago':

```
SELECT * FROM customer WHERE GRADE = 1 AND CITY <> 'Chicago';
```

```
MariaDB [Sales_db]> SELECT * FROM customer WHERE GRADE = 1 AND CITY <> 'Chicago';
+-----+-----+-----+-----+-----+
| CUSTOMER_ID | CUST_NAME | CITY | GRADE | SALESMAN_ID |
+-----+-----+-----+-----+-----+
| 20001 | Alpha Corp | New York | 1 | 10001 |
| 20004 | Delta Ltd | Houston | 1 | 10004 |
+-----+-----+-----+-----+-----+
2 rows in set (0.001 sec)
```

8. Find the minimum, maximum, and average PURCH_AMT from the orders table:

```
SELECT MIN(PURCH_AMT), MAX(PURCH_AMT), AVG(PURCH_AMT) FROM orders;
```

```
MariaDB [Sales_db]> SELECT MIN(PURCH_AMT), MAX(PURCH_AMT), AVG(PURCH_AMT) FROM orders;
+-----+-----+-----+
| MIN(PURCH_AMT) | MAX(PURCH_AMT) | AVG(PURCH_AMT) |
+-----+-----+-----+
|          200.75 |          1200.00 |          708.666667 |
+-----+-----+-----+
1 row in set (0.001 sec)
```

9. Count the number of orders placed by each customer:

```
SELECT CUSTOMER_ID, COUNT(*) AS order_count FROM orders GROUP BY CUSTOMER_ID;
```

```
MariaDB [Sales_db]> SELECT CUSTOMER_ID, COUNT(*) AS order_count FROM orders GROUP BY CUSTOMER_ID;
+-----+-----+
| CUSTOMER_ID | order_count |
+-----+-----+
|          20001 |          1 |
|          20002 |          1 |
|          20003 |          1 |
|          20004 |          1 |
|          20005 |          1 |
|          20006 |          1 |
+-----+-----+
6 rows in set (0.001 sec)
```

10. Select the top 3 highest purchase amounts from the orders table:

```
SELECT * FROM orders ORDER BY PURCH_AMT DESC LIMIT 3;
```

```
MariaDB [Sales_db]> SELECT * FROM orders ORDER BY PURCH_AMT DESC LIMIT 3;
+-----+-----+-----+-----+-----+
| ORD_NO | PURCH_AMT | ORD_DATE | CUSTOMER_ID | SALESMAN_ID |
+-----+-----+-----+-----+-----+
|    30005 |    1200.00 | 2024-05-15 |          20005 |          10005 |
|    30004 |     950.25 | 2024-04-05 |          20004 |          10004 |
|    30002 |     750.00 | 2024-02-20 |          20002 |          10002 |
+-----+-----+-----+-----+-----+
3 rows in set (0.001 sec)
```

11. Retrieve all salesmen located in 'New York' or 'Los Angeles':

```
SELECT * FROM salesman WHERE CITY IN ('New York', 'Los Angeles');
```

```
MariaDB [Sales_db]> SELECT * FROM salesman WHERE CITY IN ('New York', 'Los Angeles');
+-----+-----+-----+-----+
| SALESMAN_ID | NAME       | CITY       | COMMISSION |
+-----+-----+-----+-----+
|          10001 | John Smith | New York   |        13.50 |
|          10002 | Jane Doe   | Los Angeles |        15.00 |
+-----+-----+-----+-----+
2 rows in set (0.001 sec)
```

12. Select all orders with PURCH_AMT between 200 and 800:

```
SELECT * FROM orders WHERE PURCH_AMT BETWEEN 200 AND 800;
```

```
MariaDB [Sales_db]> SELECT * FROM orders WHERE PURCH_AMT BETWEEN 200 AND 800;
+-----+-----+-----+-----+-----+
| ORD_NO | PURCH_AMT | ORD_DATE | CUSTOMER_ID | SALESMAN_ID |
+-----+-----+-----+-----+-----+
| 30001 | 500.50 | 2024-01-15 | 20001 | 10001 |
| 30002 | 750.00 | 2024-02-20 | 20002 | 10002 |
| 30003 | 200.75 | 2024-03-10 | 20003 | 10003 |
| 30006 | 650.50 | 2024-06-20 | 20006 | 10006 |
+-----+-----+-----+-----+-----+
4 rows in set (0.001 sec)
```

13. Display all customers whose names start with 'G':

```
SELECT * FROM customer WHERE CUST_NAME LIKE 'G%';
```

```
MariaDB [Sales_db]> SELECT * FROM customer WHERE CUST_NAME LIKE 'G%';
+-----+-----+-----+-----+-----+
| CUSTOMER_ID | CUST_NAME | CITY | GRADE | SALESMAN_ID |
+-----+-----+-----+-----+-----+
| 20003 | Gamma LLC | Chicago | 3 | 10003 |
+-----+-----+-----+-----+-----+
1 row in set (0.001 sec)
```

14. Select all customers whose names contain the pattern 'Corp':

```
SELECT * FROM customer WHERE CUST_NAME LIKE '%Corp%';
```

```
MariaDB [Sales_db]> SELECT * FROM customer WHERE CUST_NAME LIKE '%Corp%'
-> ;
+-----+-----+-----+-----+-----+
| CUSTOMER_ID | CUST_NAME | CITY | GRADE | SALESMAN_ID |
+-----+-----+-----+-----+-----+
| 20001 | Alpha Corp | New York | 1 | 10001 |
+-----+-----+-----+-----+-----+
1 row in set (0.001 sec)
```

15. Select the total PURCH_AMT for each customer, but only show those customers who have total purchases over 1000

```
SELECT CUSTOMER_ID, SUM(PURCH_AMT) AS total_purchases FROM orders GROUP BY CUSTOMER_ID HAVING
total_purchases > 1000;
```

```
MariaDB [Sales_db]> SELECT CUSTOMER_ID, SUM(PURCH_AMT) AS total_purchases FROM orders GROUP BY CUSTOMER_ID HAVING total_purchases > 1000;
+-----+-----+
| CUSTOMER_ID | total_purchases |
+-----+-----+
| 20005 | 1200.00 |
+-----+-----+
1 row in set (0.001 sec)

MariaDB [Sales_db]> SELECT o.*, c.CUST_NAME FROM orders o JOIN customer c ON o.CUSTOMER_ID = c.CUSTOMER_ID;
```

16. Select all orders along with the corresponding customer names:

```
SELECT o.*, c.CUST_NAME FROM orders o JOIN customer c ON o.CUSTOMER_ID = c.CUSTOMER_ID;
```

```
MariaDB [Sales_db]> SELECT o.*, c.CUST_NAME FROM orders o JOIN customer c ON o.CUSTOMER_ID = c.CUSTOMER_ID;
+-----+-----+-----+-----+-----+-----+
| ORD_NO | PURCH_AMT | ORD_DATE | CUSTOMER_ID | SALESMAN_ID | CUST_NAME |
+-----+-----+-----+-----+-----+-----+
| 30001 | 500.50 | 2024-01-15 | 20001 | 10001 | Alpha Corp |
| 30002 | 750.00 | 2024-02-20 | 20002 | 10002 | Beta Inc |
| 30003 | 200.75 | 2024-03-10 | 20003 | 10003 | Gamma LLC |
| 30004 | 950.25 | 2024-04-05 | 20004 | 10004 | Delta Ltd |
| 30005 | 1200.00 | 2024-05-15 | 20005 | 10005 | Epsilon SA |
| 30006 | 650.50 | 2024-06-20 | 20006 | 10006 | Zeta Pty |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.001 sec)
```

17. Select all customers and their corresponding orders:

```
SELECT c.*, o.* FROM customer c LEFT JOIN orders o ON c.CUSTOMER_ID = o.CUSTOMER_ID;
```

18. Select all orders and their corresponding salesmen:

```
SELECT o.*, s.NAME, s.CITY, s.COMMISSION
```

```
FROM orders o
```

```
JOIN salesman s ON o.SALESMAN_ID = s.SALESMAN_ID;
```

```
MariaDB [Sales_db]> SELECT o.*, s.NAME, s.CITY, s.COMMISSION
-> FROM orders o
-> JOIN salesman s ON o.SALESMAN_ID = s.SALESMAN_ID;
+-----+-----+-----+-----+-----+-----+-----+
| ORD_NO | PURCH_AMT | ORD_DATE | CUSTOMER_ID | SALESMAN_ID | NAME | CITY | COMMISSION |
+-----+-----+-----+-----+-----+-----+-----+
| 30001 | 500.50 | 2024-01-15 | 20001 | 10001 | John Smith | New York | 13.50 |
| 30002 | 750.00 | 2024-02-20 | 20002 | 10002 | Jane Doe | Los Angeles | 15.00 |
| 30003 | 200.75 | 2024-03-10 | 20003 | 10003 | George Brown | Chicago | 12.75 |
| 30004 | 950.25 | 2024-04-05 | 20004 | 10004 | Nancy Davis | Houston | 11.00 |
| 30005 | 1200.00 | 2024-05-15 | 20005 | 10005 | Peter Wilson | Phoenix | 14.25 |
| 30006 | 650.50 | 2024-06-20 | 20006 | 10006 | Linda Johnson | Philadelphia | 10.50 |
+-----+-----+-----+-----+-----+-----+-----+
6 rows in set (0.001 sec)
```

19. Select all salesmen and their corresponding customers:

```
SELECT s.*, c.CUST_NAME, c.CITY AS CUSTOMER_CITY, c.GRADE
```

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FROM salesman s

JOIN customer c ON s.SALESMAN_ID = c.SALESMAN_ID;

```
MariaDB [Sales_db]> SELECT s.*, c.CUST_NAME, c.CITY AS CUSTOMER_CITY, c.GRADE  
-> FROM salesman s  
-> JOIN customer c ON s.SALESMAN_ID = c.SALESMAN_ID;
```

SALESMAN_ID	NAME	CITY	COMMISSION	CUST_NAME	CUSTOMER_CITY	GRADE
10001	John Smith	New York	13.50	Alpha Corp	New York	1
10002	Jane Doe	Los Angeles	15.00	Beta Inc	Los Angeles	2
10003	George Brown	Chicago	12.75	Gamma LLC	Chicago	3
10004	Nancy Davis	Houston	11.00	Delta Ltd	Houston	1
10005	Peter Wilson	Phoenix	14.25	Epsilon SA	Phoenix	2
10006	Linda Johnson	Philadelphia	10.50	Zeta Pty	Philadelphia	3

6 rows in set (0.001 sec)

20. Select all salesmen and their corresponding orders:

SELECT s.*, o.ORD_NO, o.PURCH_AMT, o.ORD_DATE, o.CUSTOMER_ID

FROM salesman s

JOIN orders o ON s.SALESMAN_ID = o.SALESMAN_ID;

```
MariaDB [Sales_db]> SELECT s.*, o.ORD_NO, o.PURCH_AMT, o.ORD_DATE, o.CUSTOMER_ID  
-> FROM salesman s  
-> JOIN orders o ON s.SALESMAN_ID = o.SALESMAN_ID;
```

SALESMAN_ID	NAME	CITY	COMMISSION	ORD_NO	PURCH_AMT	ORD_DATE	CUSTOMER_ID
10001	John Smith	New York	13.50	30001	500.50	2024-01-15	20001
10002	Jane Doe	Los Angeles	15.00	30002	750.00	2024-02-20	20002
10003	George Brown	Chicago	12.75	30003	200.75	2024-03-10	20003
10004	Nancy Davis	Houston	11.00	30004	950.25	2024-04-05	20004
10005	Peter Wilson	Phoenix	14.25	30005	1200.00	2024-05-15	20005
10006	Linda Johnson	Philadelphia	10.50	30006	650.50	2024-06-20	20006

6 rows in set (0.001 sec)

21. Select all salesmen from 'New York' and 'Los Angeles' using UNION:

SELECT * FROM salesman WHERE CITY = 'New York'

UNION

SELECT * FROM salesman WHERE CITY = 'Los Angeles';

```
MariaDB [Sales_db]> SELECT * FROM salesman WHERE CITY = 'New York'
-> UNION
-> SELECT * FROM salesman WHERE CITY = 'Los Angeles';
+-----+-----+-----+-----+
| SALESMAN_ID | NAME       | CITY       | COMMISSION |
+-----+-----+-----+-----+
|          10001 | John Smith | New York   |        13.50 |
|          10002 | Jane Doe   | Los Angeles |        15.00 |
+-----+-----+-----+-----+
2 rows in set (0.002 sec)
```

22. Update the COMMISSION of the salesman with SALESMAN_ID 10002 to 16.00

```
UPDATE salesman SET COMMISSION = 16.00 WHERE SALESMAN_ID = 10002;
```

23. Delete the order with ORD_NO 30003:

```
DELETE FROM orders WHERE ORD_NO = 30003;
```

24. Add a new column EMAIL to the customer table:

```
ALTER TABLE customer ADD EMAIL VARCHAR(50);
```

```
MariaDB [Sales_db]> UPDATE salesman SET COMMISSION = 16.00 WHERE SALESMAN_ID = 10002;
Query OK, 1 row affected (0.008 sec)
Rows matched: 1  Changed: 1  Warnings: 0

MariaDB [Sales_db]> DELETE FROM orders WHERE ORD_NO = 30003;
Query OK, 1 row affected (0.006 sec)

MariaDB [Sales_db]> ALTER TABLE customer ADD EMAIL VARCHAR(50);
Query OK, 0 rows affected (0.014 sec)
Records: 0  Duplicates: 0  Warnings: 0
```